SIM-01 Closure and SIM-02A Review Transmittal

Ulmo / Picket, PCK-SIM-001: SIM-01 documentation corrections and SIM-02A sensor, noise and error modeling

To	Author of the SIM-01 technical review and SIM-02 authorization
Subject	SIM-01 closure (report Revision A) and SIM-02A numerical results for review
Date	2026-11-07
Repository	`keithatulmo/picket-sim`, branch `claude/beautiful-darwin-vbeoue`
SIM-02A run of record	`20261107T214846Z_c56c0048_31da01b` (code commit `31da01b`, clean tree)
Prepared by	Claude Code (AI-generated engineering work product). Requires review by a qualified acoustic engineer before any of it informs requirements.
Requested action	Confirm SIM-01 closure; review the SIM-02A numerical results; decide the items in Section 6 before SIM-02B is authorized
Revision	A: SIM-02A report and this transmittal corrected before sending (Section 4, last group). Rev 0 package retained unchanged.

1. SIM-01 documentation corrections (review Section 1)

All five corrections are made without altering any experimental result, and recorded in ADR-0008, `results/sim01/ERRATA.md` and the report's revision history. The Rev 0 review package is retained unchanged.

#	Correction	Where
1	6.02 dB qualified as the white-noise gain of four equally weighted elements, not an absolute bound	SIM-01 report Rev A Sections 0, 4.2, 5.1; SIM-01 transmittal Rev A
2	f_HP described as a -3 dB discrimination criterion, not a limit on direction information	Report Rev A Sections 0, 4.1, 5.1 to 5.3, 8; `docs/metrics.md`
3	Median-steering bands separated from bands guaranteed across the steering set	`reports/SIM-01_band_definitions.md` (derived from the existing tables by `tools/sim01_band_definitions.py`); report Rev A Sections 0, 4.1, 5.2
4	Exact narrowband grating lobes separated from broadband direction finding	Report Rev A Sections 0, 4.1, 4.4, 5.4, 7, 8
5	Pareto conclusions conditional; no elimination of hardware candidates	Report Rev A Sections 0, 6, 7, 8

Correction 3 changes the headline band ratios. Over the horizon steering set the bands guaranteed for every direction are 1.77 (lines), 1.28 (A3) and 1.06 (A4). The median-direction bands are 3.54, 1.54 and 1.36. The Rev 0 values (3.03, 1.44, 1.18) used a mixed convention.

2. SIM-02A deliverables (review Section 5)

#	Deliverable	Status	Evidence
1	Updated technical implementation plan	Delivered	`docs/SIM-02A_plan.md`; ADR-0009 to ADR-0013 (written before implementation)
2	SIM-02A source, configuration, automated tests	Delivered	`src/picket_sim/{params,sensor,noisefields,errors,stats,rag,avs}.py`, `experiments/sim02a.py`; `configs/sim02a/`; 166 tests passing

#	Deliverable	Status	Evidence
3	Sensor, noise and error model documentation	Delivered	`docs/sensor_noise_error_models.md`
4	Comparative results, retained and new candidates	Delivered	SIM-02A report Sections 2 and 4; run tables and figures
5	Sensitivity and uncertainty analysis	Delivered	Report Sections 4.2 (eta sweep), 4.4 (Monte Carlo with 95 % intervals, one-at-a-time sweeps), 4.5 (-3/-6/-10 dB), 4.6 (tolerance scales)
6	Updated limitations and assumption register	Delivered	`docs/assumptions.md` (AS-11 to AS-19); report Section 6
7	Reproducible engineering report	Delivered	`reports/SIM-02A_engineering_report.md` / `.pdf`; `make sim02a`
8	Recommendations for SIM-02B	Delivered	Report Section 7

Additional items: acoustic vector sensor feasibility assessment (D05), `docs/avs_feasibility.md`; criterion-margin supplement (D03), `reports/SIM-02A_criterion_margin.md` (Rev A); claims check, `tools/check_sim02a_claims.py` with its output `reports/SIM-02A_claims_check.txt` (Rev A). SIM-02B work has not started.

3. Verification requirements (review Section 4)

Requirement	Evidence
Positive-semidefinite noise covariance	`check_covariance` on every coherence and total covariance (raises); `test_coherence_models_are_psd_with_unit_diagonal`, `test_invalid_covariances_and_fields_fail_visibly`
Correct noise normalization	Unit-diagonal check on every field; gain limits self-checked each run (white field 10 log10 N; isotropic field equals SIM-01 DI, agreement 5e-15 dB)
Array gain versus detection performance	Gain reported only as narrowband SNR ratios relative to A0; detection deferred to SIM-02B (ADR-0011)
Calibration and timing-error propagation	Closed forms for signal loss and null filling verified by Monte Carlo (5 error types in tests; 1,664 cases in the run, largest deviation 3.33 standard errors)
Broadband processing assumptions	None made in SIM-02A; narrowband scope stated in every result
Statistical uncertainty in Monte Carlo results	95 % intervals on means, quantiles and proportions; coverage tested; order-independent seeded streams
Consistent comparison of geometries	Footprint classes C (0.45 m) and E (0.90 m); identical steering, noise and error models for every candidate
Fail visibly	Provenance, unit, Nyquist, bandwidth, PSD, convergence and normalization violations raise; undefined statistics reported as n/a with reasons; no clipping in error paths

4. Errors found and corrected during SIM-02A

Recorded in `results/sim02a/ERRATA.md`; both earlier runs are retained unchanged.

- Run 1: line-array bearing statistics were a tie-breaking artifact. The scanned response of a line is exactly symmetric in azimuth, and `argmax` picked the mirror. Cut statistics were also reported where the error-free pattern has no half-power discrimination.
- Run 2: the first fix classified outright gross errors as ties.

- Run of record: structural ambiguity is decided on the error-free cut (outside lobe within 1 dB of the mainlobe). Gross errors are counted only where no structural ambiguity exists.
- Before any run, the assumed sensor scenario S1 was corrected to be physically consistent. Its element sensitivity had been set as if it included preamplifier gain, while the same value was used to refer the element's thermal noise.

Every table other than the Monte Carlo bearing columns is byte-identical across the three runs.

Documentation errors found by a programmatic check of every quantitative claim (`tools/check_sim02a_claims.py`) before this package was sent, corrected in report and transmittal Revision A; no run, table or figure changed:

- The circles' sidelobes were stated as "near -8 dB". The six-element circle's plateau is -6.02 dB (eight-element: -7.77 dB). That is 0.02 dB below the -6 dB criterion, so P6-C's -6 dB guaranteed band is knife-edge (`reports/SIM-02A_criterion_margin.md`).
- "At most 0.07 dB from 20 to 200 Hz" is a band median; the gain reaches 0.73 dB at 200 Hz.
- "Approach 10 log10 N" in short-correlation noise overstated the result for A1, P6-C and P8-C, which fall 0.9 to 1.7 dB short below 2 kHz.
- The 65 % gross-error rate is one of four target azimuths (range 65 to 81 %), and it comes from the combined E3 scenario, not from a 10 us timing error alone.
- Two statements were imprecise in scope: the 2 kHz host-noise range (4.7 dB is the 4 m envelope; candidates up to 1.07 m give 5.6 to 12.0 dB), and the line-array ambiguity, which has one exception at exact endfire, where target and mirror coincide.

5. Principal SIM-02A findings

1. Realized gain is governed by the spatial coherence of the dominant noise. In diffuse, ambient-limited noise, arrays up to 1.07 m gain at most 0.07 dB over A0 as a 20 to 200 Hz band median, and at most 0.73 dB at 200 Hz. With sensor self-noise 20 dB above ambient they come within 0.29 dB of 10 log10 N in every band. In short-correlation noise they come within 0.5 dB, except A1, P6-C and P8-C, which fall 0.9 to 1.7 dB short below 2 kHz.
2. Under the assumed sensors, ambient noise dominates almost everywhere. B4 is not acquirable by the low-cost scenario, and 16-bit quantization limits the mid scenario in B3 and B4.
3. At equal footprint, 6- and 8-element circles widen the -3 dB guaranteed band from 1.28 (square) to 2.53 and 3.78, but only to 1.28 and 1.32 under -10 dB, because their sidelobe plateaus sit at -6.02 and -7.77 dB. The six-element -6 dB ratio (2.37) is knife-edge: it falls to 1.65 at -6.1 dB. No other guaranteed ratio moves by more than 0.03 across +/- 0.1 dB.
4. Four-element volumetric envelopes have no guaranteed band at -6 or -10 dB (narrowband).
5. Bearing statistics are stricter than signal loss. For example, under the loose scenario E3 (1 dB, 10 deg, 1 cm, 10 us) at 15 kHz the 8-element compact circle loses 4.3 to 4.4 dB of signal but makes gross azimuth errors in 65 to 81 % of trials across the four target azimuths. Timing supplies about two thirds of E3's phase variance at 15 kHz.
6. Line arrays stay structurally ambiguous left/right under any channel errors.
7. Pareto dominance changes with the provisional tolerances; no candidate is eliminated.

All sensor, ambient, host and error levels are assumed. Ratio-based results transfer; scenario-specific numbers do not.

6. Decisions requested before SIM-02B

1. **Inputs with provenance.** Which published or measured sources to adopt for hydrophone and preamplifier self-noise, ambient noise, host noise and vibration, and flow-noise correlation (report Section 7, item 4). SIM-02B results cannot inform requirements without them.
2. **SIM-02B candidate subset and priorities** (report Section 7, item 5).
3. **Detection experiment design:** false-alarm probabilities, integration times and the signal classes to carry forward from D01.
4. **Synchronization model** for distributed modules (timing dominates at high frequency).
5. **Whether to fund the acoustic vector sensor data collection** in `docs/avs_feasibility.md`.
6. **Criterion conventions.** The structural-ambiguity level (-1 dB) used for bearing statistics and the Monte Carlo steering set (4 azimuths), if a different convention is preferred. Also whether the -6 dB criterion is applied at -6.00 dB in power or at half amplitude (-6.02 dB): P6-C's -6 dB band depends on that choice (`reports/SIM-02A_criterion_margin.md`).

7. Suggested review path (about 45 minutes)

1. This transmittal, Sections 5 and 6.
2. SIM-02A report Section 0 (findings), Figures 2, 5 and 6, and Section 7 (SIM-02B recommendations).
3. `tables/parameters_by_provenance.md`, to confirm that nothing is presented as measured.
4. SIM-01 report Rev A Section 0 and `reports/SIM-01_band_definitions.md`, for the closure corrections.
5. `reports/SIM-02A_criterion_margin.md` and `reports/SIM-02A_claims_check.txt`, for the Rev A corrections.
6. Optional: reproduce with `make sim02a` (about 6 minutes including the test suite) and compare against `SHA256SUMS.txt`; `make sim02a-claims` rechecks every quoted number in seconds, and `make sim02a-margin` regenerates the criterion-margin supplement (about 5 minutes).